Spring 2021 / EC571

Instructor:Ajay Joshi

Mid-Term Exam 2

Date: March 29, 2021

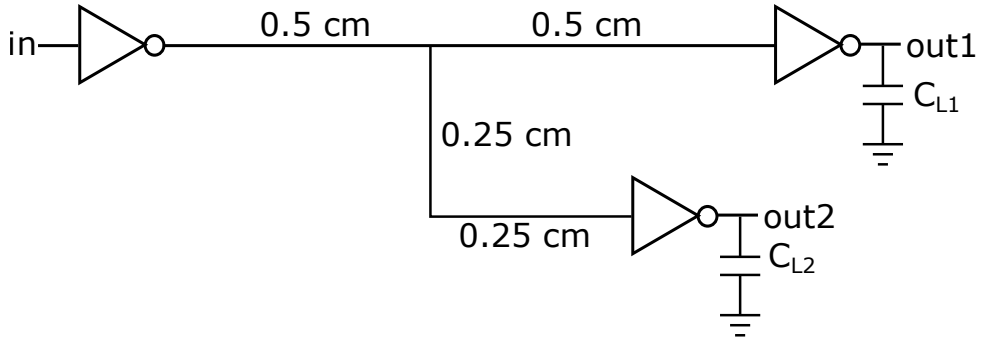
Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 1 hour 45 minutes.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
6. Material sharing is not allowed.
7. Calculators are allowed.
8. The exam is open book open notes.
9. Boston University Academic Code of Conduct applies.
10. BU ENG is collaborating with several online services, including Chegg, to monitor activity during exams and obtain account information and IP addresses connected to uploaded exam materials and their access. The penalties for cheating are extremely severe including expulsion from BU.

Question 1



1. For the circuit shown in the above figure, draw the equivalent RC network that includes all the parts. Assume a lumped model for each interconnect segment. (6 points)

2. Write the Elmore delay equations for the entire circuit from in to out1, and from in and out2. (8 points)

3. Calculate the Elmore delay for out1 and out2. You can use the following technology parameters for calculating the delay - V_{dd} = 1.5 V, L = 0.25 micron $R_{sq(NMOS)}$ = 10 Kohm/sq, $R_{sq(PMOS)}$ = 25 Kohm/sq, C_G = 1 fF/micron, C_D = 1.5 fF/micron, wire capacitance = 0.25 fF/micron, and wire resistance = 15 ohms/micron. Assume that all inverters use a minimum width of 300 nm for the NMOS. You will need to choose the width of the PMOS transistor such that the low-to-high delay of the inverters matches with the high-to-low delay. Assume a C_{L1} = 20 fF and C_{L2} = 10 fF. (8 points)

Question 2

Consider the circuit shown below. Assume a unit clk-to-q delay for the register, two units of delay for the multiplexer, one unit of dely for demultiplexer, and two units of delay for each logic block. The registers are positive-edge triggered and have a setup time of one unit.

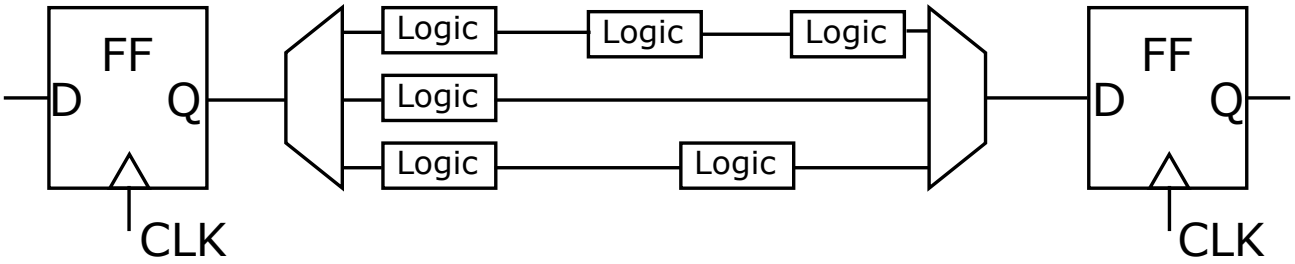


Figure 1

1. Determine the minimum clock period for the above circuit. (7 points)

2. Assuming a hold time of 2 time unit, what is the maximum positive clock skew that can be tolerated by this circuit? (7 points)

3. If the clock input at both FFs has a jitter of ± 1 delay unit, what would be the minimum clock period of the circuit? Assume zero clock skew. (6 points)

Question 3

Sketch the schematics for $f(P,Q,R,S) = \overline{P \cdot Q + R \cdot S}$ using each of the following circuit techniques. You can assume that you have both complemented and non-complemented inputs:

1. Static CMOS. (5 points)

2. Pseudo-nMOS. (5 points)

3. Dynamic. (5 points)

4. Pass transistors (including the inverter at the end). (5 points)

Question 4

1. In your own words, explain the concepts of negative setup time and negative hold time (4 points)

2. In your own words, explain how clock-to-q delay of a flip-flop increases as we start pushing the input data edge closer to the clock edge. You can use the flip-flop circuit that we discussed in class to answer this question. (4 points)

3. Consider the circuit shown below where we have a wire driven by an inverter and there is a same sized inverter on the receiver side i.e. at the other end of wire.

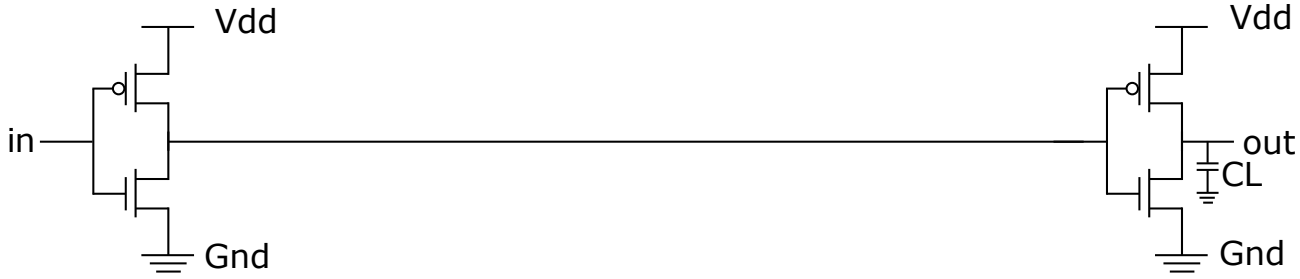


Figure 2

(a) If the wire width is doubled, then what will happen to the delay of the overall circuit? Consider both scenarios - one where wire R and C parasitics dominate the R and C parasitics of the transistors in the two inverters. Use Elmore delay equation for the circuit to support your argument. (3 points)

(b) If the wire thickness is doubled, then what will happen to the delay of the overall circuit? Consider both scenarios - one where wire R and C parasitics dominate the R and C parasitics of the transistors in the two inverters. Use Elmore delay equation for the circuit to support your argument. (3 points)

(c) If the wire length is halved, then what will happen to the delay of the overall circuit? Consider both scenarios - one where wire R and C parasitics dominate the R and C parasitics of the transistors in the two inverters. Use Elmore delay equation for the circuit to support your argument. (3 points)

(d) If the oxide thickness (between wire and substrate) is doubled, then what will happen to the delay of the overall circuit? Consider both scenarios - one where wire R and C parasitics dominate the R and C parasitics of the transistors in the two inverters. Use Elmore delay equation for the circuit to support your argument. (3 points)

Question 5

For the circuit shown in Figure 3, the signal waveforms per cycle at nodes A, B, C, D and E are shown next to it. You can assume the these signal waveforms repeat every cycle 'T'. You can assume the following device parameters: V_T = 0.3 V, R_{ON} = 10 kΩ and R_{OFF} = infinity, V_{dd} = 1.0 V.

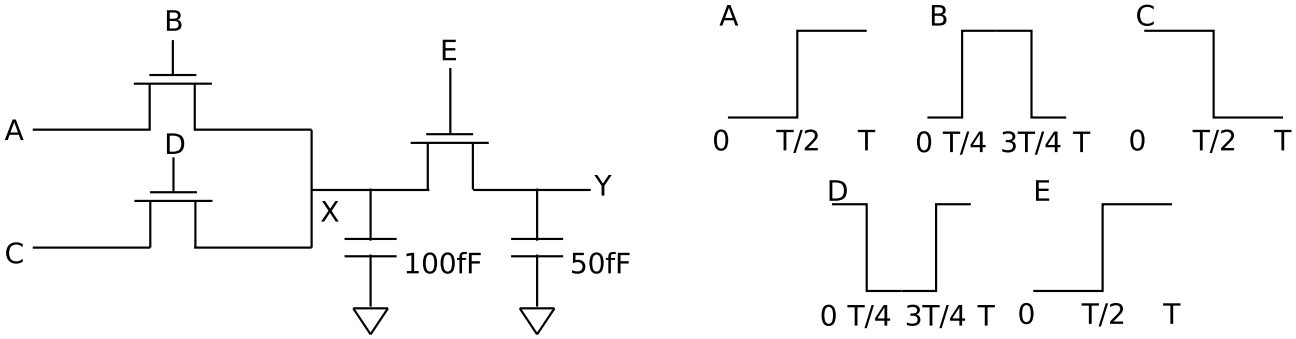


Figure 3

1. Draw the timing waveforms of the signals on nodes X and Y for a time of 0 to 4T. You can assume the signals at nodes A, B, C, D and E are switching between 0 and 1.0 V. Assume no charge sharing. (8 points)

2. Calculate the total dynamic power dissipated during each of the 4 cycles by the capacitors are nodes X and Y. (8 points)

3. Draw the timing waveforms of the signals on nodes X and Y for a time of 0 to 4T if all the NMOS transistors were replaced with transmission gates. You can assume the signals at nodes A, B, C, D and E are switching as shown in Figure 3, and between 0 and 1.0 V. (4 points)

Question 6

1. Figure 4 shows the schematic of a modified Switched Output Differential Structure (SODS) logic. Explain how the logic style operates. Comment on the strengths and weaknesses of this design style as compared to the static complementary CMOS technology in terms of power, performance, area, and reliability. You can assume that both NAND and AND have 2 inputs and you have access to A, B, A' and B' (10 points).

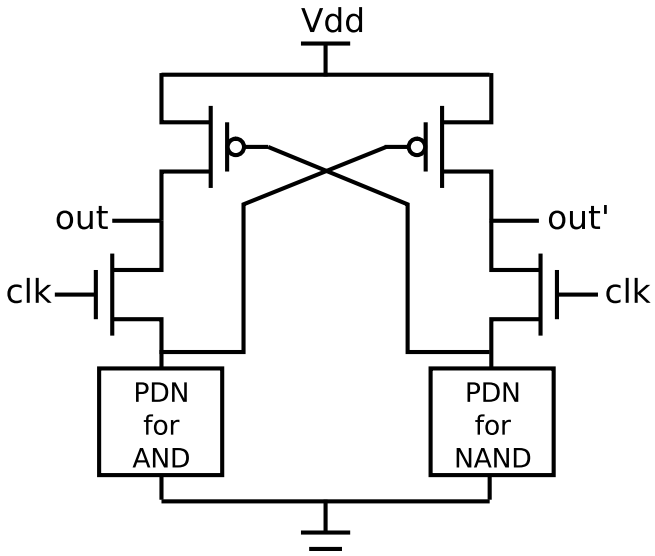


Figure 4

2. In your own words, explain the working of a negative edge-triggered flip-flop and a positive level-triggered latch. (5 points)

3. Design a negative edge-triggered flip-flop using two latches. (5 points)

Bonus question

Prof. Joshi is thinking of having a barbecue party at his place on Patriots day. Given that he is a vegetarian, suggest five food items that he can cook on a barbecue grill. (5 points)